The empirical recurrence for OEIS A237094: recovering a conjecture that is not written down

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Abstract

OEIS A237094 records "Empirical recurrence of order 68", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 198$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237094 is "Number of (n+1)X(3+1) 0..2 arrays with the maximum plus the lower median minus the upper median of every 2X2 subblock equal.". It has offset 1 and begins

$$1178, 10466, 95954, 944850, 9188374, 93666930, 938845280, 9750633426, \ \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:50:24, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 198$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 198$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 396$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{68} c_i\, a(n-i),$$

c_1	17	c_{18}	−5705819538177700	c_{35}	6951652824081120069687	c_{52}	962879210235
c_2	246	c_{19}	−9967837639920746	c_{36}	22015507529076587192103	c_{53}	−741727864381
c_3	−5453	c_{20}	76291965835562923	c_{37}	−17533211295329157209281	c_{54}	−31381657405
c_4	−20221	c_{21}	78448968992360697	c_{38}	−41387261651180890895209	c_{55}	222217206838
c_5	759750	c_{22}	−796376235533403485	c_{39}	34987038385868104577034	c_{56}	788314842601
c_6	124313	c_{23}	−412226597198155528	c_{40}	63498605174627120138651	c_{57}	−49888346746
c_7	−61530020	c_{24}	6569791232939372644	c_{41}	−55672201254443713992647	c_{58}	−15002529382
c_8	100314902	c_{25}	944274938263804063	c_{42}	−79384411489641742008977	c_{59}	81622948360
c_9	3259430748	c_{26}	−43233959166514534743	c_{43}	70850674594978526817476	c_{60}	21149775200
c_{10}	−9028838267	c_{27}	5873541459574841176	c_{44}	80633779554436532846676	c_{61}	−9350096336
c_{11}	−120197775916	c_{28}	228604546976883816538	c_{45}	−72082276240779681619331	c_{62}	−2137488807
c_{12}	436715652097	c_{29}	−79517628553437492667	c_{46}	−66255934065978664239440	c_{63}	7042704466
c_{13}	3200046315784	c_{30}	−976753675648474583409	c_{47}	58436609033184108714792	c_{64}	1465324776
c_{14}	−14094302199658	c_{31}	497902805474839887926	c_{48}	43782377273123551790532	c_{65}	−311044872
c_{15}	−62822802565580	c_{32}	3386901726650349179950	c_{49}	−37530439381567458811888	c_{66}	−61045137
c_{16}	327620230151098	c_{33}	−2145422168622445107031	c_{50}	−23091251239166534865856	c_{67}	606981918
c_{17}	918313354889994	c_{34}	−9560842780586283056943	c_{51}	18933758814135478875344	c_{68}	116429606

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68+4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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